

Certificate No.: 3161791-001
Serial No.: 3161791
Model: 919P-010-16
Part No.: 774059A
Description: Power and Energy Sensor

Calibrated at:
Calibration Lab
Ophir Optronics Solutions
10 Hartom Street
Jerusalem, Israel
Tel: +972-2-548-7407
Email: service.ophir.il@mksinst.com

Measurement Uncertainty

1.9%

Standards Used for Calibration

Model	F150A-BB-SM KM7350-01	F30A-BB-SM-SM KM7348-01	F150A-BB-SM KM7358-01		
Serial No.	795613	795616	798616		
Next Cal. Date	18-APR-2025	17-APR-2025	18-APR-2025		

Calibration Measurements

Power (W)	Master		Sensor Under Test			Results		
Wavelength	Scale	Reading	Scale	Reading	λ Setting	Deviation	Tolerance	Pass/Fail
1066 nm	30 W	4.848 W	10 W	4.860 W	1066	0.25 %	± 3.0 %	PASS
532 nm	5 W	1.9625 W	5 W	1.9679 W	532	0.28 %	± 3.0 %	PASS
10600 nm	30 W	4.747 W	10 W	4.760 W	10600	0.27 %	± 3.0 %	PASS

Energy (J)	Reference		Sensor Under Test		Results		
λ Setting	Calculated Energy		Energy	Scale	Deviation	Tolerance	Pass/Fail
532	1.591 J		1.612 J	2 J	1.32 %	± 5.0 %	PASS

Notes:

Outgoing Result: PASS

Calibration Procedure: QA-103

Performed By: Yossi Yatzkan - Q.A. 42

Calibration Date: 12-JAN-2025

Temperature: 22 °C

Issue Date: 12-JAN-2025

Recommended Recal. Date: JUL 2026

Humidity: 39 % RH

For data sheets of all standard Newport products, please go to the relevant product page at: www.newport.com

The results reported on this certificate apply only to the item calibrated at the specified calibration points. Conformity to product specification is determined by a binary guard band decision rule for normal distributions as defined by JCGM 106:2012 with at least 96 % confidence for measurement results at acceptance limit. The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor $k=2$, which for a normal distribution corresponds to a coverage probability of approximately 95 %.

For information about uncertainty of measurement, please refer to: <http://www.ophiropt.com/laser--measurement/knowledge-center/article/1110>

Recommended calibration intervals and the resulting due dates are 18 months for new products and 12 months for recalibrated products. The calibration interval can be changed by the customer at their own discretion, but Ophir Optronics will not guarantee the calibration after the stated intervals. All standards used are traceable to NIST and/or other national and international standards. The entire traceability chart and certifications are available on request.

The Israel Laboratory accreditation authority (ISRAC) is one of the signatories of the International Accreditation Cooperation (ILAC) arrangement for the mutual recognition of testing results. The use of ISRAC symbol relates to tests/calibrations which are included in the organization scope of accreditation and performed according to the accreditation rules as detailed in the accreditation certificate. ISRAC is not responsible for the testing results conducted by the laboratory and the laboratory's accreditation is not considered as an approval of ISRAC or a different party related to the assessed. This document must be referred to entirely and copying of any part of it to other documents is forbidden.

End of certificate

China RoHS Declaration

部件名称	有害物质					
	铅 (Pb)	汞 (Hg)	镉 (Cd)	六价铬 (Cr(VI))	多溴联苯 (PBB)	多溴联苯醚 (PBDE)
774059A	X	0	0	0	0	0

此表是按照 SJ/T 11364 的规定编制。

0: 表示该部件的所有均质材料中所含该有害物质均低于 GB/T 26572 的规定。

X: 表示该部件所使用的至少一种均质材料中所含该有害物质高于 GB/T 26572 的规定。

(企业可根据实际情况在空格中对“X”标记进一步提供技术说明。)

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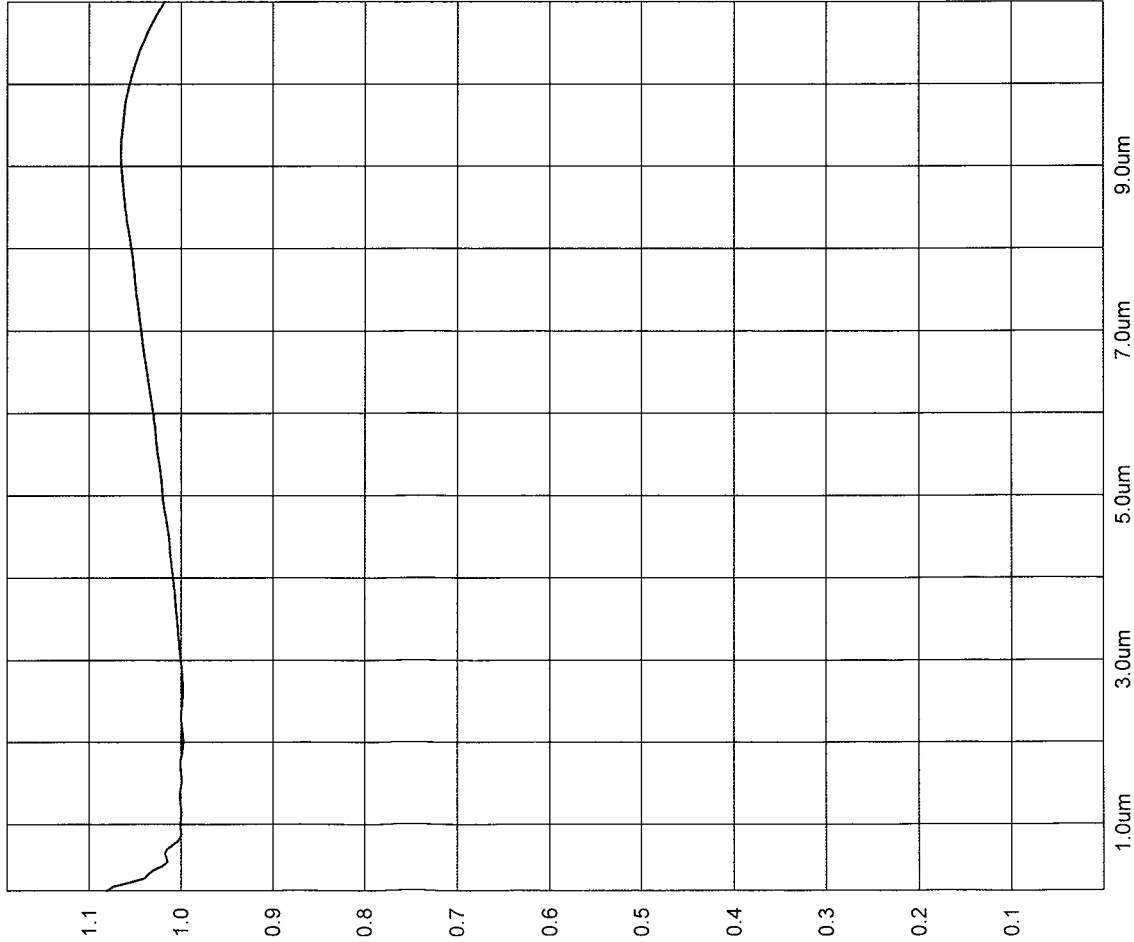
www.newport.com

WAVELENGTH SENSITIVITY PLOT

Model Name: 919P-010-16
Serial No: 3161791

Spectrophotometer: Perkin Elmer 950

SENSITIVITY relative to 1064nm



SENSITIVITY RELATIVE TO 1064nm

W/L(nm) SENS	W/L(nm) SENS	W/L(nm) SENS	W/L(nm) SENS	W/L(nm) SENS	W/L(nm) SENS
190 1.081	2190 0.999	4190 1.011	6190 1.033	8190 1.057	10190 1.051
240 1.074	2240 1.000	4240 1.011	6240 1.034	8240 1.058	10240 1.050
290 1.055	2290 1.000	4290 1.012	6290 1.034	8290 1.058	10290 1.049
340 1.039	2340 1.000	4340 1.012	6340 1.035	8340 1.059	10340 1.047
390 1.036	2390 0.999	4390 1.013	6390 1.036	8390 1.060	10390 1.046
440 1.030	2440 0.999	4440 1.013	6440 1.036	8440 1.060	10440 1.044
490 1.020	2490 0.998	4490 1.013	6490 1.037	8490 1.061	10490 1.042
540 1.015	2540 0.998	4540 1.014	6540 1.038	8540 1.061	10540 1.040
590 1.016	2590 0.998	4590 1.015	6590 1.038	8590 1.062	10590 1.038
640 1.018	2640 0.998	4640 1.015	6640 1.039	8640 1.062	10640 1.036
690 1.016	2690 0.998	4690 1.016	6690 1.040	8690 1.063	10690 1.034
740 1.010	2740 0.998	4740 1.017	6740 1.041	8740 1.063	10740 1.031
790 1.004	2790 0.998	4790 1.018	6790 1.041	8790 1.063	10790 1.029
840 1.000	2840 0.999	4840 1.018	6840 1.042	8840 1.064	10840 1.027
890 1.000	2890 0.999	4890 1.019	6890 1.042	8890 1.064	10890 1.024
940 1.000	2940 1.000	4940 1.020	6940 1.043	8940 1.065	10940 1.021
990 1.000	2990 1.000	4990 1.020	6990 1.043	8990 1.065	10990 1.019
1040 1.000	3040 1.001	5040 1.021	7040 1.044	9040 1.065	11000 1.018
1090 1.000	3090 1.001	5090 1.021	7090 1.044	9090 1.065	
1140 0.999	3140 1.001	5140 1.021	7140 1.045	9140 1.065	
1190 1.000	3190 1.002	5190 1.022	7190 1.046	9190 1.065	
1240 1.000	3240 1.002	5240 1.022	7240 1.046	9240 1.065	
1290 1.001	3290 1.003	5290 1.023	7290 1.047	9290 1.065	
1340 1.001	3340 1.003	5340 1.023	7340 1.047	9340 1.065	
1390 1.001	3390 1.004	5390 1.024	7390 1.048	9390 1.064	
1440 1.000	3440 1.004	5440 1.025	7440 1.049	9440 1.064	
1490 0.999	3490 1.004	5490 1.025	7490 1.049	9490 1.063	
1540 0.999	3540 1.005	5540 1.026	7540 1.050	9540 1.063	
1590 1.000	3590 1.005	5590 1.026	7590 1.050	9590 1.062	
1640 1.000	3640 1.006	5640 1.027	7640 1.050	9640 1.062	
1690 1.000	3690 1.006	5690 1.027	7690 1.051	9690 1.062	
1740 1.000	3740 1.006	5740 1.028	7740 1.051	9740 1.061	
1790 1.000	3790 1.007	5790 1.028	7790 1.052	9790 1.060	
1840 0.999	3840 1.007	5840 1.029	7840 1.052	9840 1.060	
1890 0.998	3890 1.008	5890 1.029	7890 1.053	9890 1.059	
1940 0.998	3940 1.008	5940 1.030	7940 1.053	9940 1.057	
1990 0.997	3990 1.009	5990 1.030	7990 1.054	9990 1.056	
2040 0.997	4040 1.009	6040 1.031	8040 1.055	10040 1.055	
2090 0.998	4090 1.010	6090 1.032	8090 1.055	10090 1.054	
2140 0.999	4140 1.010	6140 1.032	8140 1.056	10140 1.053	

Sensitivity at 1064nm: 56.399nAW

Note: This sensor is calibrated over the region 190-11000nm, and at the following laser points:
532nm, 1066nm, 10600nm;

See 'Certificate Of Calibration' and Datasheet for more details.